

SIMATS ENGINEERING ASSESSMENT TOOL 2(MEDIUM)

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO3: Analyze network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Port Scanning Challenge

1. Using Nmap, perform a security assessment on the server 192.168.1.20. Identify all open ports and running services. Analyze the purpose of each discovered service and recommend security measures to reduce the attack surface of the system.
2. Using Nikto, assess the web server hosted on 192.168.1.30. Identify potential web server misconfigurations, outdated components, or exposed files and directories. Discuss the security risks associated with your findings and suggest appropriate mitigation strategies.
3. Using Netcat (nc), investigate the host 192.168.1.40 by connecting to open ports and collecting available service banners. Analyze the information obtained and explain how attackers could use banner information during the reconnaissance phase of an attack.
4. Using Telnet, examine the server 192.168.1.50 to determine whether remote administration services are accessible. Identify any security concerns related to the discovered services and recommend more secure alternatives where appropriate.
5. Using Netdiscover, perform host discovery on the network segment 192.168.1.1–50. Identify active hosts on the network and analyze how the discovered information could assist an attacker in planning further enumeration or exploitation activities..

Code of Conduct:

K Nitheesh

I (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K Nitheesh

CO3 – ASSESSMENT TOOL 2 (MEDIUM)

PORT SCANNING CHALLENGE – DETAILED ANSWERS

Student Name: K Nitheesh Kumar

Register Number: 192325090

Course Name: Ethical Hacking | Course Code: ITA14

Course Outcome Covered: CO3 – Analyze network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. The assessment carries 30% weightage, with a duration of 90 minutes and total marks of 50. 穀filecite學turn4file0學L3-L10像

Source coverage: The supplied assessment contains five practical questions covering Nmap, Nikto, Netcat, Telnet and Netdiscover. The detailed answers below follow the same order and terminology as the source. 穀filecite學turn4file0學L11-L27像

Important evidence note: The PDF provides target addresses and required tools but does not provide actual scan outputs. Therefore, the report explains the correct procedure, interpretation framework and expected security analysis. Actual observed ports, banners and active hosts should be inserted from the authorized laboratory run rather than invented.

QUESTION 1 – NMAP SECURITY ASSESSMENT OF 192.168.1.20

1.1 Requirement and Objective

The first task requires an Nmap-based security assessment of server 192.168.1.20. The assessment must identify all open ports and running services, explain the purpose of each service, and recommend measures that reduce the system's attack surface. 穀filecite學turn4file0學L11-L13傢

1.2 Host Discovery

Host discovery is the first stage because it establishes whether the server is reachable before the tester performs deeper enumeration. In an authorized lab, the tester can use ICMP-based discovery and Nmap's host-discovery mode.

```
ping -c 4 192.168.1.20
nmap -sn 192.168.1.20
```

A successful ping suggests reachability, but an unsuccessful ping does not necessarily mean that the host is offline. Firewalls may block ICMP while still allowing TCP services. Therefore, host discovery should be interpreted together with subsequent port-scanning results.

1.3 TCP Port Scanning

```
nmap -sS 192.168.1.20
nmap -p- 192.168.1.20
```

The SYN scan identifies TCP ports that appear open, closed or filtered. A full-range scan with -p- checks all TCP ports instead of relying only on Nmap's default set. The final report should record the exact port number, protocol, service and version returned by the authorized server.

1.4 Service and Version Detection

```
nmap -sV 192.168.1.20
nmap -sS -sV 192.168.1.20
```

Version detection is important because the same port can host different applications. For example, TCP 80 commonly carries HTTP, but the actual software may be Apache, Nginx, IIS or another server. Version information enables the tester to compare the installed component with supported versions and security advisories.

1.5 OS Fingerprinting and Scripts

```
sudo nmap -O 192.168.1.20
nmap -sC 192.168.1.20
sudo nmap -sS -sV -O -sC 192.168.1.20 -oN Q1_nmap.txt
```

OS fingerprinting provides an estimate of the operating system based on network behavior. Default scripts collect additional service information. The saved normal-output file provides reproducible evidence for the report.

1.6 Port and Service Analysis Framework

Observed Item	Purpose to Record	Security Question
Port number	TCP/UDP port and state	Why is this port reachable?
Service	Application/protocol	Is the service required for the server role?
Version	Software/version detected	Is the component supported and patched?
Exposure	Internal/external reachability	Who needs access to it?
Authentication	Whether credentials are required	Is authentication strong and protected?
Risk	Likelihood × impact	What happens if the service is compromised?

1.7 Attack-Surface Interpretation

An open port is not automatically a vulnerability. The correct assessment compares the technical exposure with the server's business purpose. A public web server may legitimately require HTTPS, whereas an administrative database, remote-management service or legacy protocol may not need public exposure. Services that are not required should be disabled or restricted at the firewall.

1.8 Recommendations

- Create an inventory of all listening services and assign a documented business owner.
- Disable services that are not required for the server's role.
- Restrict administration to a management network or VPN.
- Patch and upgrade exposed services.
- Use strong authentication and MFA for privileged access where supported.
- Apply host-based firewall rules using least privilege.
- Reduce unnecessary service/version disclosure.
- Monitor authentication attempts and unusual connection patterns.
- Repeat the Nmap assessment after remediation.

QUESTION 1 – THREAT-ACTOR USE OF ENUMERATION

1.9 How Enumeration Supports an Attack

The source specifically asks how a threat actor could use enumeration information for service exploitation, brute-force attacks and vulnerability discovery. 穀filecite學turn4file0學L14-L18傢 Enumeration reduces uncertainty: once an attacker knows that a particular service and version are reachable, they can prioritize research and testing against that service instead of probing blindly.

Enumeration Information	Possible Attacker Use	Defensive Response
Open port	Identify accessible network function.	Close unnecessary ports.
Service name	Select protocol-specific tests.	Restrict exposure and monitor access.
Software version	Research known vulnerabilities.	Patch and remove obsolete software.
OS fingerprint	Choose platform-specific techniques.	Harden OS and minimize fingerprintable exposure.
Authentication service	Attempt credential attacks.	Use MFA, rate limits and network restrictions.
Web service	Map application attack surface.	Secure application, WAF and monitoring.
Banner information	Correlate technology stack.	Minimize unnecessary banners.

1.10 Brute-Force Risk

If enumeration identifies a remote authentication service such as SSH or another login interface, an attacker may prioritize credential attacks. The security control is not simply to hide the port. The organization should reduce exposure, enforce strong passwords, implement MFA where possible, apply rate limiting/lockout controls carefully, and monitor failed authentication.

1.11 Vulnerability-Discovery Risk

Service-version detection can be compared with vendor security advisories and vulnerability databases. If an obsolete version is found, the tester should verify whether the vulnerability is applicable to the exact configuration before reporting it as confirmed. A version match alone is evidence for investigation, not necessarily proof of exploitability.

1.12 Risk-Rating Model

Finding Type	Likelihood	Impact	Typical Rating
Unnecessary public service	Medium	Medium	Medium
Outdated exposed service	High	High	High
Public administrative interface	Medium–High	High	High
Weak authentication service	Medium–High	High	High
Minor banner disclosure	High	Low	Low

1.13 Final Answer for Q1

The recommended assessment is to discover the host, scan the full TCP range, detect services and versions, fingerprint the OS and run appropriate default scripts. Every open service should be justified against the server's intended function. Unnecessary services should be disabled, administrative services should be restricted, exposed components should be patched, and monitoring should be enabled. Enumeration information can help attackers select service-specific exploitation, credential attacks and vulnerability research, so reducing information exposure and attack surface is an important defensive objective.

QUESTION 2 – NIKTO ASSESSMENT OF WEB SERVER 192.168.1.30

2.1 Requirement and Objective

The second task requires Nikto assessment of the web server at 192.168.1.30. The assessment must identify potential web-server misconfigurations, outdated components, exposed files and directories, then explain the risks and recommend mitigation strategies. 穀filecite學turn4file0學L14-L17像

2.2 Initial Web-Service Verification

```
curl -I http://192.168.1.30
```

```
curl -i http://192.168.1.30
```

The response should be reviewed for status code, Server header, redirects and other security-relevant headers. If HTTPS is available, the tester should also inspect the HTTPS endpoint.

```
curl -I https://192.168.1.30
```

2.3 Nikto Scan

```
nikto -h http://192.168.1.30
```

```
nikto -h http://192.168.1.30 -output Q2_nikto.txt
```

Nikto is useful for identifying common web-server configuration problems, known risky files, potentially outdated components and other observable issues. Findings must be manually validated because automated tools can produce false positives.

2.4 Misconfiguration Analysis

Possible Finding	Security Risk	Mitigation
Directory listing enabled	May expose files and application structure.	Disable listing unless explicitly required.
Default files/pages	Reveal software or configuration information.	Remove unnecessary default content.
Backup files	May expose source code or credentials.	Remove from web root and protect backups.
Verbose server banner	Assists fingerprinting.	Minimize unnecessary version disclosure.
Missing security headers	Weakens browser-side defenses.	Configure suitable headers.
Outdated server component	May contain known vulnerabilities.	Patch or upgrade.
Unnecessary HTTP methods	Can expand attack surface.	Disable methods not required.

2.5 Exposed Files and Directories

If Nikto reports a file or directory, the tester should verify whether it is actually accessible and whether its content is sensitive. A directory name alone is not proof of a vulnerability. Sensitive configuration files, backup archives, source-code files and administrative interfaces deserve higher priority because they can expose credentials, application logic or privileged functionality.

2.6 Risk Evaluation

Risk depends on both exposure and consequence. A low-value banner disclosure may be Low risk, while an exposed configuration file containing database credentials may be High or Critical. An outdated web server with a confirmed remotely exploitable vulnerability should receive immediate attention.

2.7 Mitigation Strategy

- Patch the web server and underlying operating system.
- Remove unused modules and default content.
- Disable directory listing where not required.

- Remove backup archives and configuration files from publicly served directories.
- Restrict administrative paths with authentication and network controls.
- Configure HTTPS and secure redirects.
- Set appropriate security headers such as HSTS and Content-Security-Policy according to application requirements.
- Use secure file permissions and keep secrets outside the web root.
- Review Nikto findings manually and document confirmed results.
- Repeat the scan after remediation.

QUESTION 2 – DETAILED WEB-SERVER RISK ANALYSIS

2.8 Outdated Components

An outdated component can increase risk because security fixes may not be present. The correct process is to record the detected version, identify the supported release, check whether the installation is still maintained, and verify whether relevant vulnerabilities apply. The organization should upgrade to a supported version and test application compatibility before deployment.

2.9 Exposed Administrative Resources

Administrative interfaces should not be publicly reachable unless there is a clear business requirement and strong controls. A preferred architecture places management functions behind a VPN, management VLAN, allowlist or other restricted path. MFA and strong audit logging further reduce risk.

2.10 HTTP Response and Header Review

```
curl -s -D - http://192.168.1.30/ -o /dev/null
```

Review headers such as Server, Location, Strict-Transport-Security, Content-Security-Policy, X-Content-Type-Options and framing controls where applicable. Missing headers are generally defense-in-depth findings; severity should reflect the actual application risk rather than assuming every missing header is a critical vulnerability.

2.11 Finding Register Template

ID	Finding	Evidence	Severity	Recommended Action
Q2-F1	Outdated component	Nikto/version evidence	High if vulnerable	Upgrade and retest
Q2-F2	Exposed sensitive file	HTTP response/path evidence	High	Remove/restrict and rotate secrets
Q2-F3	Directory listing	Browser/Curl evidence	Medium	Disable listing
Q2-F4	Verbose banner	Response header	Low	Reduce disclosure
Q2-F5	Missing security header	HTTPS response	Low/Medium	Configure appropriate header

2.12 Final Answer for Q2

Nikto should be used to identify likely web-server weaknesses, but each finding should be validated manually. The security team should prioritize exposed sensitive files, outdated exploitable components and unrestricted administrative interfaces over minor information-disclosure observations. The strongest mitigation is a combination of patch management, minimal web-root content, restricted administration, secure headers, HTTPS, least privilege and continuous monitoring.

QUESTION 3 – NETCAT BANNER ENUMERATION OF 192.168.1.40

3.1 Requirement and Objective

The third task requires Netcat (nc) to investigate 192.168.1.40 by connecting to open ports and collecting available service banners. The assessment must analyze the information obtained and explain how attackers could use banners during reconnaissance. 穀filecite學turn4file0學L18-L21傢

3.2 Discover Candidate Ports

Netcat is most useful after ports have been identified. In an authorized environment, Nmap can establish which ports are open and Netcat can then connect to selected services for banner observation.

```
nmap -sS -sV 192.168.1.40
```

3.3 Netcat Connections

```
nc -nv 192.168.1.40 22
```

```
nc -nv 192.168.1.40 80
```

```
nc -nv 192.168.1.40 25
```

The -n option avoids DNS resolution and -v provides verbose connection information. Some services immediately return a banner, while others require a protocol-specific request. For an HTTP service, a simple request can be used in a controlled lab:

```
printf 'HEAD / HTTP/1.0\r\nHost: 192.168.1.40\r\n\r\n' | nc 192.168.1.40 80
```

3.4 Banner Information

Banner Element	What It May Reveal	Security Relevance
Service name	SSH/HTTP/SMTP/etc.	Identifies protocol and attack surface.
Software name	Apache/OpenSSH/etc.	Enables technology fingerprinting.
Version	Specific release	Can be compared with security advisories.
Operating-system clues	Platform-specific behavior	Helps select later tests.
Host/domain information	Naming conventions	Supports infrastructure mapping.
Protocol features	Supported capabilities	May reveal legacy/unsafe configuration.

3.5 Attacker Use of Banners

During reconnaissance, an attacker can correlate banners with known weaknesses, determine which services deserve attention and avoid wasting effort on irrelevant ports. A precise software version may allow a threat actor to search for publicly documented vulnerabilities. This is why unnecessary banner disclosure should be minimized, although hiding a banner is not a substitute for patching.

3.6 Defensive Recommendations

- Minimize unnecessary software/version disclosure.
- Keep all exposed services patched.
- Restrict administrative services by network location.
- Use secure protocols such as SSH instead of Telnet.
- Monitor repeated connection attempts and unusual protocol behavior.
- Use firewalls to expose only required services.
- Validate that banners do not reveal internal hostnames, paths or configuration details.

QUESTION 3 – BANNER ANALYSIS AND RISK ASSESSMENT

3.7 Example Interpretation

Suppose a controlled lab connection to port 22 returns an SSH banner identifying a specific OpenSSH release. The correct conclusion is that SSH is present and the software version can be investigated for support status. The tester should not immediately state that the host is vulnerable. The next step is to compare the version with vendor advisories and verify configuration and authentication controls.

3.8 Banner-to-Attack-Path Linkage

Step	Attacker Question	Defensive Question
1	Which ports are reachable?	Which ports are actually required?
2	Which services run there?	Are these services authorized?
3	What software/version is used?	Is it supported and patched?
4	What protocol features are exposed?	Can unnecessary features be disabled?
5	What later attack is plausible?	What control breaks that attack path?

3.9 Evidence Documentation

A strong report should include the exact Netcat command, target port, date/time, sanitized banner output, interpretation and risk. Credentials, tokens or private data must not be included in the report.

3.10 Final Answer for Q3

Netcat provides a simple way to observe service banners after open ports have been identified. Banner information can reveal service type, software family, version and other implementation details. Attackers can use those details to prioritize vulnerability research and service-specific attacks. Defenders should therefore minimize unnecessary disclosure, patch services, restrict exposure and monitor reconnaissance patterns. The key principle is that information hiding alone is not sufficient; the service itself must be securely configured.

QUESTION 4 – TELNET EXAMINATION OF 192.168.1.50

4.1 Requirement and Objective

The fourth task requires Telnet to examine server 192.168.1.50 and determine whether remote administration services are accessible. It also asks for security concerns and secure alternatives.

4.2 Telnet Connectivity Test

```
telnet 192.168.1.50 23
```

If TCP port 23 is reachable, the connection may display a Telnet login prompt or service banner. The presence of Telnet is a significant concern because Telnet was designed without modern encryption for protecting credentials and interactive sessions.

4.3 Testing Other Remote-Administration Ports

The tester should identify authorized remote-management services using Nmap before connecting with Telnet. Examples include SSH on TCP 22, RDP on TCP 3389 and other management services depending on the server platform.

```
nmap -sV -p 22,23,3389 192.168.1.50
```

4.4 Security Concerns

Issue	Explanation	Risk
Plaintext credentials	Telnet does not provide modern encrypted transport for login credentials.	High/Critical
Session interception	Network observers may capture interactive traffic.	High
Legacy protocol	Telnet is generally unsuitable for secure administration.	High
Internet exposure	A public management port increases attack surface.	High
Weak authentication	Credential compromise can lead to remote access.	High/Critical
Poor logging	Legacy systems may have weaker centralized monitoring.	Medium/High

4.5 Secure Alternatives

Need	Preferred Alternative	Security Benefit
Secure shell administration	SSH	Encrypted administration and strong authentication options.
Windows remote desktop	RDP with NLA + MFA/VPN where appropriate	Stronger controlled remote access.
Secure web management	HTTPS	Encrypts management traffic.
Network monitoring	SNMPv3	Authentication and privacy for management traffic.

4.6 Hardening Recommendations

- Disable Telnet completely when it is not required.
- If legacy Telnet is temporarily unavoidable in an isolated lab, restrict it to that isolated network and do not use sensitive credentials.
- Use SSH for Linux/Unix administration.
- Restrict management ports with firewall rules and VPNs.

- Enable MFA for privileged remote access where supported.
- Disable default/unused administrator accounts.
- Use strong unique passwords and monitor failed login attempts.
- Patch the server and remote-management software.
- Centralize logs and alert on repeated authentication failures.

QUESTION 4 – REMOTE-ADMINISTRATION RISK ANALYSIS

4.7 Least-Privilege Administration

Remote administration should be available only to personnel and systems that require it. A secure architecture separates ordinary user traffic from the management plane. Administrators should connect through a trusted management subnet or VPN rather than exposing management services directly to an untrusted network.

4.8 Attack Scenario

An attacker may first discover TCP 23, identify the Telnet service and observe its banner. If the service is reachable, the attacker may attempt to identify valid credentials through unauthorized credential attacks. If credentials are compromised, the attacker could obtain interactive access. Because Telnet traffic is not appropriately protected, interception can also create an opportunity for credential exposure on networks where an attacker can observe traffic.

4.9 Risk Reduction

Control	Effect
Disable Telnet	Removes insecure remote-administration exposure.
SSH	Provides encrypted remote shell access.
VPN	Places administration behind a protected access boundary.
MFA	Reduces reliance on passwords alone.
Firewall allowlisting	Limits management access to approved sources.
Central logging	Improves detection and investigation.
Patch management	Reduces software vulnerability exposure.

4.10 Final Answer for Q4

The Telnet test should determine whether TCP 23 and other management services are accessible. If Telnet is present, it should be treated as a high-risk legacy protocol because it does not provide the protections expected from modern secure administration. The preferred response is to disable Telnet, use SSH or another secure management protocol, restrict administrative access through a VPN or management network, require strong authentication, and monitor administrative activity.

QUESTION 5 – NETDISCOVER HOST DISCOVERY: 192.168.1.1–50

5.1 Requirement and Objective

The fifth task requires Netdiscover to perform host discovery over the network segment 192.168.1.1–50. The objective is to identify active hosts and analyze how that information could help an attacker plan further enumeration or exploitation. 穀filecite學turn4file0學L25-L27像

5.2 Network Range

The stated range represents the addresses from 192.168.1.1 through 192.168.1.50. In an Ethernet laboratory network, ARP-based discovery can identify devices that respond to ARP requests. The exact active-host list must come from the authorized lab execution.

5.3 Netdiscover Command

```
sudo netdiscover -r 192.168.1.0/26
```

Because the question specifies .1–50 rather than an entire subnet, the report should document the exact addresses observed and avoid treating unused addresses as active. If the laboratory network is actually a /24 or another subnet, the correct range should be confirmed before scanning.

5.4 Active-Host Information

Information	Meaning	Security Value
IP address	Network-layer address of responding host.	Identifies potential targets.
MAC address	Hardware/interface address.	Helps identify device/vendor.
Vendor	Manufacturer inferred from MAC OUI.	May indicate device type.
Response behavior	Whether host responds to discovery.	Shows current reachability.
Host role	Inferred from additional authorized enumeration.	Helps prioritize testing.

5.5 How Attackers Use Host Discovery

An attacker can use host discovery to build an inventory of the network before selecting specific systems for port scanning. Knowing which addresses are active helps focus reconnaissance on likely servers, workstations, gateways and infrastructure devices. MAC vendor information can provide clues about device type. This information can support later service enumeration and attack planning.

5.6 Defensive Measures

- Use network segmentation to limit visibility between unrelated security zones.
- Place sensitive systems in protected VLANs.
- Use NAC to control which devices may connect.
- Monitor ARP anomalies and unauthorized devices.
- Maintain an accurate asset inventory so unknown devices can be investigated.
- Restrict management interfaces to dedicated networks.
- Use firewalls to prevent unnecessary east-west traffic.

QUESTION 5 – HOST-DISCOVERY ANALYSIS AND ATTACK-SURFACE PLANNING

5.7 Host Inventory Template

Host IP	MAC Address	Vendor	Likely Role	Follow-Up	Risk
192.168.1.x	Record from Netdiscover	Record from output	Server/workstation/gateway	Nmap service scan	To be assessed
192.168.1.x	Record from Netdiscover	Record from output	Network device	Management-service review	To be assessed
192.168.1.x	Record from Netdiscover	Record from output	Workstation	Limited service enumeration	To be assessed

5.8 Linkage Between Discovery and Exploitation

Host discovery itself does not exploit a system. Its security significance comes from the information it provides for later stages. An attacker may use the discovered inventory to select a gateway for management-service testing, a server for service enumeration or a workstation for lateral-movement research. Defenders should therefore treat unauthorized network discovery as an early indicator and correlate it with subsequent port scans.

5.9 Detection and Response

- Monitor network traffic for unusual ARP sweeps and repeated discovery patterns.
- Alert when an unmanaged device appears on a protected network.
- Correlate discovery activity with firewall, IDS and endpoint telemetry.
- Investigate unknown MAC addresses against the asset inventory.
- Segment high-value systems so discovery of one segment does not expose the entire organization.
- Use NAC and switch security controls where appropriate.

5.10 Final Answer for Q5

Netdiscover can identify active hosts in the specified laboratory segment by observing ARP responses. The exact active hosts must be recorded from the authorized scan. The resulting IP/MAC/vendor information helps create a network inventory and can guide subsequent enumeration. From a defensive perspective, segmentation, NAC, asset inventory, monitoring and restricted management access reduce the value of such reconnaissance.

INTEGRATED PORT-SCANNING ANALYSIS

Relationship Between the Five Tools

Tool	Primary Function	Result	Next Security Step
Nmap	Port/service/OS enumeration	Open ports, services, versions	Risk assessment and focused testing
Nikto	Web-server assessment	Web misconfigurations and risky resources	Manual validation and remediation
Netcat	Banner/service observation	Raw service responses/banners	Technology fingerprinting
Telnet	Remote-service connectivity check	Login/banner behavior	Replace insecure management protocols
Netdiscover	Host discovery	Active IP/MAC/vendor information	Network inventory and focused enumeration

Defense-in-Depth Strategy

The five exercises illustrate a complete reconnaissance-to-hardening workflow. Netdiscover identifies active hosts, Nmap identifies their exposed services, Netcat provides service-level banner evidence, Nikto evaluates the web-server layer, and Telnet demonstrates the risks of legacy administration. The outputs should be linked rather than treated as isolated tool results.

Priority Model

Priority	Typical Finding	Recommended Action
Critical	Insecure management protocol or exploitable exposed service	Disable/restrict immediately and remediate.
High	Outdated public-facing service, sensitive exposed resource	Patch/remove/restrict and retest.
Medium	Unnecessary but limited-risk service	Disable or restrict during hardening cycle.
Low	Minor banner/configuration disclosure	Reduce exposure as defense in depth.

Professional Documentation

For every tool, record the target, command, date/time, relevant output, interpretation, risk and remediation. Screenshots should show the command and important result while hiding passwords, tokens and private information. The supplied rubric emphasizes identification, analysis, linkage, application and clarity; therefore, the report should explicitly connect each scan result to a security implication and practical control.

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FINAL CONCLUSION

Port scanning and enumeration are foundational activities in ethical hacking because they establish what a system exposes before deeper vulnerability assessment begins. The first case demonstrates how Nmap creates a structured inventory of ports, services, versions and operating-system characteristics. The second case extends this process to the web layer using Nikto, where configuration weaknesses, outdated components and exposed resources can be identified. The third case shows how Netcat can provide raw service-banner evidence that helps interpret what is actually running. The fourth case highlights the security problems created by legacy remote administration through Telnet. The fifth case demonstrates how host discovery creates a network-level inventory that can guide later enumeration.

Across all five cases, the central security principle is attack-surface reduction. Every exposed service should have a documented purpose, should be reachable only by the users or systems that require it, and should be maintained at a supported security level. Management services deserve special protection because compromise can provide privileged access. Web services require secure configuration and regular vulnerability assessment. Network segmentation limits the consequences of a single compromised host.

The correct professional workflow is therefore: authorize the scope → discover hosts → identify ports → detect services and versions → analyze business necessity → validate findings → assign risk → remediate → retest. This approach supports CO3 because it applies port scanning, ping sweeps and enumeration techniques to different operating-system and network-service contexts. 穀filecite學turn4file0學L5-L7傢

Submission Checklist

- K Nitheesh Kumar / 192325090 entered on the report.
- Original assessment page retained.
- Nmap commands and findings documented for 192.168.1.20.
- Nikto methodology and risk analysis documented for 192.168.1.30.
- Netcat banner methodology documented for 192.168.1.40.
- Telnet security assessment documented for 192.168.1.50.
- Netdiscover methodology documented for 192.168.1.1–50.
- Risk levels and mitigation strategies included.
- Actual laboratory outputs/screenshots to be inserted where required.
- Sensitive information removed from evidence.

Student Declaration

I K Nitheesh Kumar (192325090) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Nitheesh Kumar

Register Number: 192325090

Q.No	Identification of Port Scanning Factors(25)	Analysis(25)	Linkage(20)	Application(20)	Clarity(10)	Marks
1						
2						
3						
4						
5						

Signature of Faculty

Total Avg :

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Identification of Port Scanning Factors	25%	Clearly identifies comprehensive and relevant strengths, weaknesses, opportunities, and threats	Identifies most factors with minor gaps	Limited or partially relevant identification	Incorrect or incomplete Port Scanning factors
Analysis	25%	Provides deep, insightful analysis of each Port Scanning component with strong reasoning	Provides good analysis with some depth	Basic analysis with limited reasoning	No meaningful analysis provided
Linkage	20%	Effectively links Port Scanning elements to derive strong strategic insights and implications	Shows some linkage with moderate insights	Weak linkage; limited strategic thinking	No linkage or strategic insight
Application	20%	Applies Port Scanning analysis effectively to the given case or scenario with clear relevance	Applies analysis with minor gaps	Limited application; lacks clarity	No clear application to context
Clarity	10%	Well-structured, clear, and logically presented with proper formatting	Generally clear with minor issues	Some structure but lacks clarity	Poorly organized and difficult to understand